15/9/24 Experiment - 7

Aim:
Write a program to implement flow control data link layer using sliding window protocol. Simulate the flow of frames from one node to another

code:

```
import time
message = input("Enter the text message:")
window_size = int(input("Enter window size:"))
error_frame = int(input("Enter error frame(0 for
                no error):"))
if window_size <= 0:
    print("Invalid window size")
    exit()
print("\n Sliding window protocol")
print("\n Message:", message)
print("Window Size:", window_size)
start = 0
expected_frame = 1
error_induced = False
while start < len(message):
    end = min(start + window_size, len(message))
    print("\n""Sender Buffer")
    Sender_buffer = []
    for i in range(start, end):
        frame_no = i+1
        data = message[i]
        Sender_buffer.append([frame_no, data])
        print("Frame_No=", frame_no, "DATA=", data)
print("\n Sending frames")
time.sleep(1)
```

```
print("\n Receiver ")
error_found = False
for frame in qsender_buffer:
    frame_no = frame[0]
    data = frame[1]
    if frame_no == error_frame and not
                                error_induced:
        received_frame = frame_no + 1
        error_induced = True
    else:
        received_frame = frame_no
    print(
      "Recieved Frame:",
          recived_frame,
          "DATA:", data)
    if recived_frame == expected_frame:
        print("Frame", reaived_frame, "is Correct")
        expected_frame += 1
    else:
      print("\n Error detected")
      print("Expected Frame:", expected_frame)
      print("Recieved Frame:", received_frame)
      print("Reciever Buffer")
      print("NACK =", expected_frame)
      print("\n NACK recieved by Sender")
      print("Resending frame from frame", expected_frame
      error_found = True
      break
  if error_found:
      qstart = expected_frame - 1
      time.qsleep(1)
      Continue
print("\n Reciver Buffer")
ack = expected_frame
```

```
print("Ack = ", ack)
print("\n Ack received by Sender")
print(" Expected ACK", end+1)
if ack == end+1
    print("\n Ack is correct")
    print(" Sliding window moves forward")
    start = end
else:
    print("\n Incorrect Ack")
    print(" Waiting for correct Ack").
print("\n Orginal Message:", message)
print(" All frames successfully transmitted")
```

Input

Enter text message: HELLO

Enter window size: 3

Enter Error frame (0 for no error): 2

o/p:

message: HELLO

Window size: 3

Frame_no = 1 Data H

Frame_no = 2 Data = E

Frame_no = 3 Data = L

Sending Frame

Recived Frame : 3 Data: E

Error detected:

Expected Frame: 2

Recieved Frame: 3

NAck = 2

NAck Recivid by sender

Resending Frames from frame 2
Frame_no = 2 data = E
Frame_no = 3 data = L
Frame_no = 4 Data = L.

Sending frames...
Recieved frame: 2 Data: E
Frame 2 is correct
Recieved Frame: 3 Data: C
Frame 3 is correct
Received Frame: 4 Data: L
Frame 4 is correct.

Ack = 5
Ack received by sender
Expected Ack: 5
Ack is correct
Recived Frame: 5 Data = O
Ack = 6
Ack received by sender
Expected Ack: 6
Ack is correct
original message: HELLO
All frames successfully transmitted.

Result:
program to implement sliding window has been executed successfully.